SMALL CLASS 6: Numerical integration with spreadsheets SOLUTIONS

In this class, you will use a spreadsheet to implement 3 numerical methods: right-Riemann sums and the trapezoidal method (from last small class) and Simpson's rule (new, this class), and bound the error for the latter two.

Small class questions

1. **(DEMO) Simpson's rule.** With the help of your instructor, write down the formula for approximating an integral using Simpson's rule. Why must n be an even number?

   Simpson's rule:

   $$\int_a^b f(x)\,\mathrm{d}x \approx \frac{\Delta x}{3}\left(f(x_0)+4f(x_1)+2f(x_2)+4f(x_3)+2f(x_4)+\ldots+4f(x_{n-1})+f(x_n)\right).$$

   We need n to be an even number because Simpson's rule approximates each segment of a function as a parabola, which requires 3 points. Three points span two sub-intervals.

2. **Implementing numerical integration methods with spreadsheets.**

   Our goal is to approximate $\int_0^2 \sin\left(x^2\right)\,\mathrm{d}x \approx 0.804776489\ldots$ using all three of our numerical methods via spreadsheet. Note that there is *no* technique that will allow us to integrate this exactly.

   (a) For the numerical schemes that follow, we will take $n = 20$. Compute Δx.

   The bounds are from 0 to 2, and we want $n = 20$ intervals: $\Delta x = \dfrac{b-a}{n} = \dfrac{2-0}{20} = \dfrac{1}{10} = 0.1$.

   (b) Memory check: Can any or all of your team members recall the formulas for our other two numerical schemes? Discuss, and write them below. (*If you don't have them memorized, see them on the next page and write them out here – and discuss how you might remember them!*)

   You should write them out here, to practice – but for the purposes of this solution document, they are listed below.

   Right-Riemann sum: $\displaystyle\int_a^b f(x)\,\mathrm{d}x \approx \sum_{i=1}^{n} f(a+i\Delta x)\Delta x = \Delta x\left(f(x_1)+f(x_2)+\ldots f(x_n)\right)$

   Trapezoidal method: $\displaystyle\int_a^b f(x)\,\mathrm{d}x \approx \frac{\Delta x}{2}\left(f(x_0)+2f(x_1)+2f(x_2)+\ldots+2f(x_{n-1})+f(x_n)\right)$

   Simpson's: $\displaystyle\int_a^b f(x)\,\mathrm{d}x \approx \frac{\Delta x}{3}\left(f(x_0)+4f(x_1)+2f(x_2)+4f(x_3)+2f(x_4)+\ldots+4f(x_{n-1})+f(x_n)\right)$

   (c) Open a spreadsheet (Excel, LibreOffice Calc, Google Sheets). Label, then fill in, the following columns:

A	B	C	D	E	F	G	H	I
i	x_i	f(x_i)	R-coeff	T-coeff	S-coeff	RRiemann	Trapezoid	Simpson

   - `A`: this is your counter i. It starts at 0 and goes up to $n = 20$.
   - `B`: this is your x_i. Recall $x_i = a + i\Delta x$, where the cell in column `A` holds your i.
   - `C`: $f(x_i)$. You can use the cell in column `B` together with the built-in function `SIN()`.
   - `D, E, F`: these are the coefficients (the number that multiplies $f(x_i)$) for the three methods: right-Riemann sum, trapezoidal rule, and Simpson's rule, respectively. You can enter these by hand but try finding a faster way (dragging the pattern down is great).
   - `G, H, I`: these are the coefficients for your respective method multiplied by the function value.

   Once you have all the columns set up, you can use the `SUM()` function on columns `G, H, I`, and apply the correct multiplication to find your desired value.

   See a sample completed spreadsheet here; you can click on any cell to see its formula.
   Results: Right-Riemann = 0.7647491705; Trapezoidal = 0.8025892953; Simpson's = 0.8048108161.

(d) Recall that $\int_0^2 \sin\left(x^2\right)\,\mathrm{d}x \approx 0.804776489\ldots$ was given to us (presumably computed by someone with a very good computational method). Compare your three approximations to this actual value. Which method performs better? Is that what you expected? Discuss.

Given $\int_0^2 \sin\left(x^2\right)\,\mathrm{d}x \approx 0.804776489\ldots$, the errors are calculated as the absolute value of the difference between the approximate value and the actual value: Right-Riemann: 0.040027319; Trapezoidal: 0.002187194; Simpson's: 0.000034327. Simpson's method gives the closest approximation. This is expected, as Simpson's rule uses parabolas to estimate the function, rather than lines (Trapezoidal) or constant values (Riemann).

The next parts (e)(f) of this question are *not* on the spreadsheet.

The maximum error in approximating $\int_a^b f(x)\,\mathrm{d}x$:

Assume that $|f''(x)| \le M$ for all x in $[a,b]$. Then the total error introduced by the Trapezoidal method is bounded by $\frac{M}{12}\frac{(b-a)^3}{n^2}$. Further, if $|f^{(4)}(x)| \le L$ for all x in $[a,b]$, then the total error introduced by Simpson's rule is bounded by $\frac{L}{180}\frac{(b-a)^5}{n^4}$.

If you need these error formulas on an exam, they will be provided to you. Reminder: Check the Math 101 Textbook for learning objectives.

(e) What is the maximum error in your computation of $\int_0^2 \sin\left(x^2\right)\,\mathrm{d}x$ using the Trapezoidal method? You may use the fact that $f''(x) = 2\cos\left(x^2\right) - 4x^2\sin\left(x^2\right)$. *Hint:* The triangle inequality states that $|a+b| \le |a| + |b|$.

How does the error bound compare to the actual error?

The first step in computing the Trapezoidal error bound is calculating a value M such that $|f''(x)| = |2\cos\left(x^2\right) - 4x^2\sin\left(x^2\right)| \le M$ for all x in $[0,2]$. The triangle inequality tells us that $|2\cos\left(x^2\right) - 4x^2\sin\left(x^2\right)| \le |2\cos\left(x^2\right)| + |4x^2\sin\left(x^2\right)|$. The properties of sin and cos tell us that $|\cos\left(x^2\right)| \le 1$ and $|\sin\left(x^2\right)| \le 1$, and we know that x^2 attains its maximum on $[0,2]$ at $x = 2$, where $x^2 = 2^2 = 4$. Put this all together in one clean computation:

$$\begin{aligned} |f''(x)| &= |2\cos\left(x^2\right) - 4x^2\sin\left(x^2\right)| \\ &\le |2\cos\left(x^2\right)| + |4x^2\sin\left(x^2\right)| && \text{using the triangle inequality} \\ &\le |2| + |4\cdot 2^2| && \text{considering function maxima} \\ &= 18. \end{aligned}$$

Thus, set $M = 18$. Then the total error introduced by the Trapezoidal method is less than or equal to

$$\frac{M}{12}\frac{(b-a)^3}{n^2} = \frac{18}{12}\frac{(2-0)^3}{20^2} = \frac{3}{10^2} = 0.03.$$

Indeed, our approximation is an order of magnitude more accurate.

(f) **Challenge.** What is the maximum error in your computation of $\int_0^2 \sin\left(x^2\right)\,\mathrm{d}x$ using Simpsons rule? You may use the fact that $f^{(4)}(x) = 4(4x^4 - 3)\sin\left(x^2\right) - 48x^2\cos\left(x^2\right)$.

How does the error bound compare to the actual error?

The first step in computing the error bound for a Simpson's rule approximation is calculating a value L such that $|f^{(4)}(x)| = |4(4x^4-3)\sin(x^2) - 48x^2\cos(x^2)| \leq L$ for all x in $[0,2]$. Although this is a bit more involved than the previous calculation, it follows the same ideas. We can use the triangle inequality again, which tells us that $|4(4x^4-3)\sin(x^2) - 48x^2\cos(x^2)| \leq |4(4x^4-3)\sin(x^2)| + |48x^2\cos(x^2)|$. The properties of sin and cos tell us that $|\cos(x^2)| \leq 1$ and $|\sin(x^2)| \leq 1$. Finally, we know that $4x^4 - 3$ attains its maximum on $[0,2]$ at $x = 2$, where $4x^4 - 3 = 4 \cdot 2^4 - 3$, and similarly $x^2 \leq 2^2$ on the same interval. Put this all together in one clean, careful computation, along with a little bit of patience and careful simplification:

$$\begin{aligned}
|f^{(4)}(x)| &= |4(4x^4 - 3)\sin(x^2) - 48x^2\cos(x^2)| \\
&\leq |4(4x^4 - 3)\sin(x^2)| + |48x^2\cos(x^2)| && \text{using the triangle inequality} \\
&\leq |4(4 \cdot 2^4 - 3)| + |48 \cdot 2^2| && \text{considering function maxima} \\
&\leq |2^2(2^6 - 3)| + |3 \cdot 2^4 \cdot 2^2| = 2^2 \cdot 109.
\end{aligned}$$

This suggests we take $L = 2^2 \cdot 109$, which enables us to bound the total error introduced by Simpson's rule by

$$\frac{L}{180}\frac{(b-a)^5}{n^4} = \frac{2^2 \cdot 109}{180}\frac{(2-0)^5}{20^4} = \frac{109 \cdot 2^5}{45 \cdot 20^4} = \frac{109}{2^3 \cdot 3^2 \cdot 5^5} = \frac{109}{225000} = 0.00048444444.$$

Again, our approximation is an order of magnitude more accurate, well within this bound.

3. **Extra: Using GeoGebra to approximate the integral of an function that is not defined explicitly.** Given a definite integral where we know the *shape* of the function but not the *formula* (or the formula is not a standard function) – we can use numerical integration to compute approximations to the integral. We can do this by drawing a function and approximating the area underneath it using an online calculator called `GeoGebra`.

   (a) In GeoGebra (`https://www.geogebra.org/calculator`):

   - Click on "Tools". Scroll down to click on "MORE". Scroll down to click on "Freehand Shape".
   - Draw a freehand function (any curve) that spans across the interval $[-5, 5]$. Make sure your curve goes well past -5 and 5.
   - Click on "Algebra" to see that your function is now defined as f, even with no explicit formula.

See a sample completed geogebra computation here. Note that this example uses the function g instead of f.

   (b) Use the "Algebra" tab to enter an equation to approximate the integral of your function from -5 to 5 using the Trapezoidal method with $n = 5$. Recall

$$\int_a^b f(x)\,dx \approx \frac{\Delta x}{2}\left(f(x_0) + 2f(x_1) + 2f(x_2) + \ldots + 2f(x_{n-1}) + f(x_n)\right).$$

Calculate $\Delta x = \dfrac{b-a}{n} = \dfrac{5-(-5)}{5} = 2$, and fill in the missing pieces of the formula:

$$\int_{-5}^{5} f(x)\,dx \approx \underbrace{\frac{2}{2}\left(f(-5) + 2f(-3) + 2f(-1) + 2f(1) + 2f(3) + f(5)\right)}_{\text{type this into a Geogebra Input.}}.$$

(c) **Extra Challenge.** Write the formula in (b) in summation notation. (Hint: you'll have to pull out the first and last term.)

$$\int_{-5}^{5} f(x)\ \mathrm{d}x \approx \frac{2}{2}\left(f(-5) \quad + \quad 2\sum_{i=1}^{4} f(-5+2i) \quad + f(5) \right)$$

To type a sum in Geogebra, you can use the "Sum" function:

$$\sum_{i=A}^{N} f(i) = \mathrm{Sum}\left(f(i), i, A, N\right),$$

where the first entry is what to sum, the second is the counter, the third is the start, the fourth is the end. Type a new Geogebra input that approximates $\int_{-5}^{5} f(x)\ \mathrm{d}x$ using the Trapezoidal method with $n = 5$, that uses the "Sum" function.

$$\frac{2}{2}(f(-5) + 2\mathrm{Sum}\left(f(-5+2i), i, 1, 4\right) + f(5))$$

(You don't need to type the $\frac{2}{2}$, as this is 1; we show it here just to make the factor of $\frac{\Delta x}{2}$ visible.)

(d) **Extra Challenge.** Make a new input that approximates $\int_{-5}^{5} f(x)\ \mathrm{d}x$ using the Trapezoidal method with $n = 20$ that uses the "Sum" function.

If $n = 20$, then $\Delta x = \frac{5-(-5)}{20} = 0.5$. So all our Δx terms will be 0.5 instead of 2, and our sum now includes all of the middle terms: 18 of them, from $i = 1$ to $i = 19$:

$$\frac{0.5}{2}(f(-5) + 2\mathrm{Sum}\left(f(-5+0.5i), i, 1, 19\right) + f(5))$$

This was not asked, but: to generalize this one step further, we could write the trapezoidal formula in terms of any function f defined over an interval from a to b, and consider the areas of n sub-intervals of width $\Delta x = D$ defined as $\frac{b-a}{n}$:

$$\int_{a}^{b} f(x)\ \mathrm{d}x \approx \frac{D}{2}\left(f(a) + 2\sum_{i=1}^{n-1} f(a+Di) + f(b) \right).$$

In GeoGebra, we would have to define n, a, and b as numbers, and $D = \frac{b-a}{n}$, and draw a function f, and then type:

$$\frac{D}{2}(f(a) + 2\mathrm{Sum}\left(f(a+Di), i, 1, n-1\right) + f(b))$$

Practice questions Solutions not provided; see some answers available in text, or ask on piazza.

The questions below are for practice. They do not contribute to your grade, and it is not expected that you complete them during your small class. However, you are strongly encouraged to work through them as they provide practice for this week's large and small classes.

4. Inspired by Math 101 Textbook, Example 1.12.2: Use a spreadsheet to approximate $\int_{0}^{1} \frac{4}{1+x^2}\ \mathrm{d}x$, using all 3 rules, with 8 intervals. (Check your answer for the trapezoidal rule against the example.)
5. Inspired by Math 101 Textbook, Example 1.12.3: Use a spreadsheet to approximate $\int_{0}^{\pi} \sin x\ \mathrm{d}x$, using all 3 rules, with 8 intervals. (Check your answer for the trapezoidal rule against the example.)
6. Numerical Integration: Math 101 Problem Book, Section 1.12: Q1, Q2, Q3-Q6, Q8, Q9, Q11-Q20, Q25-Q30

7. Improper integrals: Math 101 Problem Book, Section 1.13: Q1-Q4, Q6-Q13, Q16, Q18, Q19, Q23-Q25.

8. Consider $\int_{-\infty}^{\infty} x \, \mathrm{d}x$. Does this integral converge or diverge? Consider $\lim_{T\to\infty} \int_{-T}^{T} x \, \mathrm{d}x$. Does this limit exist and if so what is its value? Explain why there is no contradiction here.